

Mobile Development

Hello World from Flutter !

LAB 3/W4

3RD YEAR - ENSIA

OBJECTIVES

- Develop Hello World App in Flutter
- Build Beautiful User Interfaces in Flutter

TASKS BEFORE THE LAB

- Flutter/Dart must be setup using VS Code
- Develop **“Hello World”** App using Flutter where you need to use the Widget Column / Card / Text showing the text “Hello from Ensia, my name is”.
- The Hello World App must be running on your real Phone

STRICTLY INDIVIDUAL EXERCISES (NOT TEAMS)

Task / Exercise C1 (Using Functions for Getting Widgets) [60 mins] :

Part 1 : Creating Widget Component:

1. Create a class named : **PricingScreen** as a **stateful Widget** which shows an appBar title and a header Text.
2. Within the state class, Create a Flutter **function** that **returns** a Widget. The function should be named **getPriceBoxWidget** that takes three arguments:
 1. header : Top Text
 2. description: Bottom Text
 3. is_selected

The returned widget should show a rounded box as shown on the right.

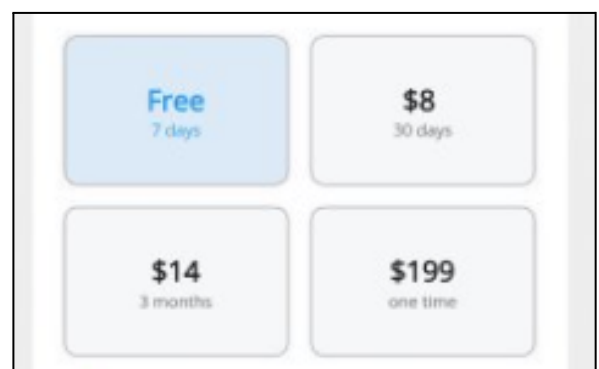
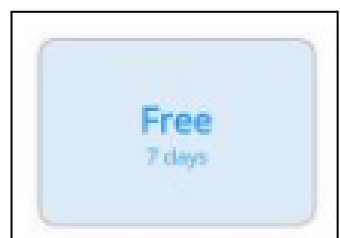
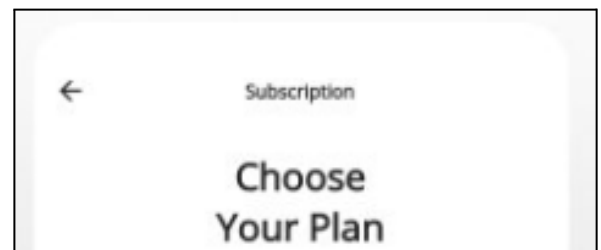
The button box is when the argument isSelected is given as True.

Part 2 : Constructing data

1. Within the state class, Initialize a List that contains four Map elements in the form :

```
[  
  {  
    'header':'Free',  
    'description':'7 days',  
    'is_selected':True  
  },  
  ..  
]
```

2. Write the flutter code within the build function that draws the four boxes as shown on the right.
 - You invoke the global function using the initialized data.



Part 3 : Completing the Full UI

Complete the Flutter screen to draw the following screen.

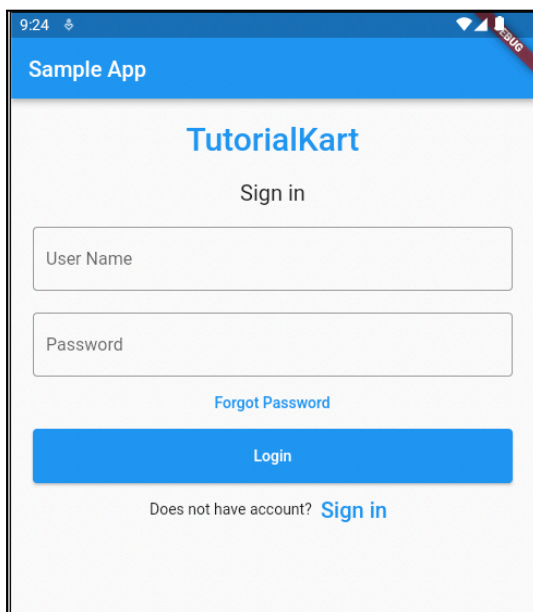
The circled check icon is : `check_circle`

Task / Exercise C2 (Design simple UI in Flutter) [45 mins + 15 mins class demonstration] :

For the following screen:

- Write down the tree of widgets in form:
 - MaterialApp:
 - Scaffold
 - Column...
- Using VS, write the flutter code for it.
 - **Make sure to create a new screen file called: login.dart where you create a stateless widget for the login screen.**

(No need to implement any interaction or business logic.)

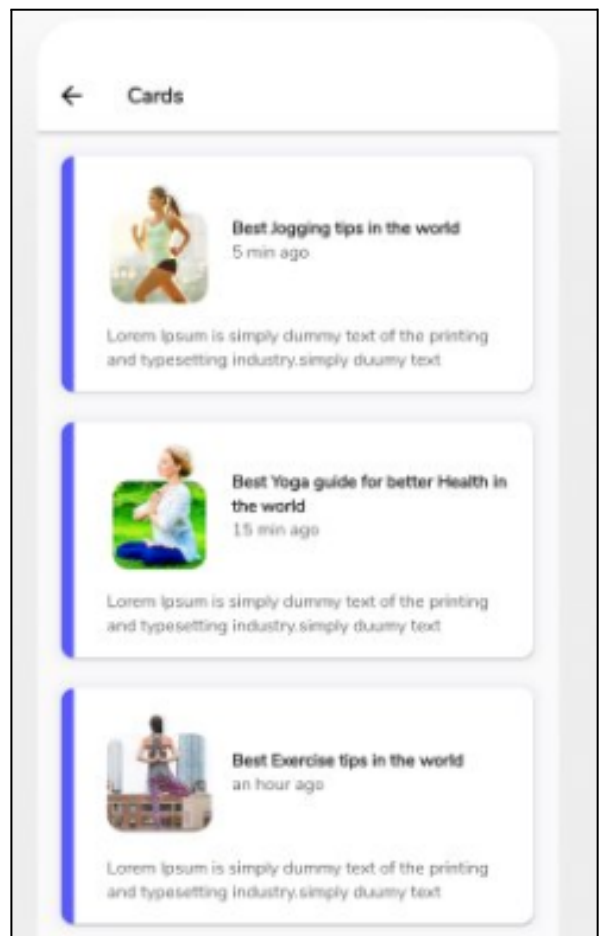
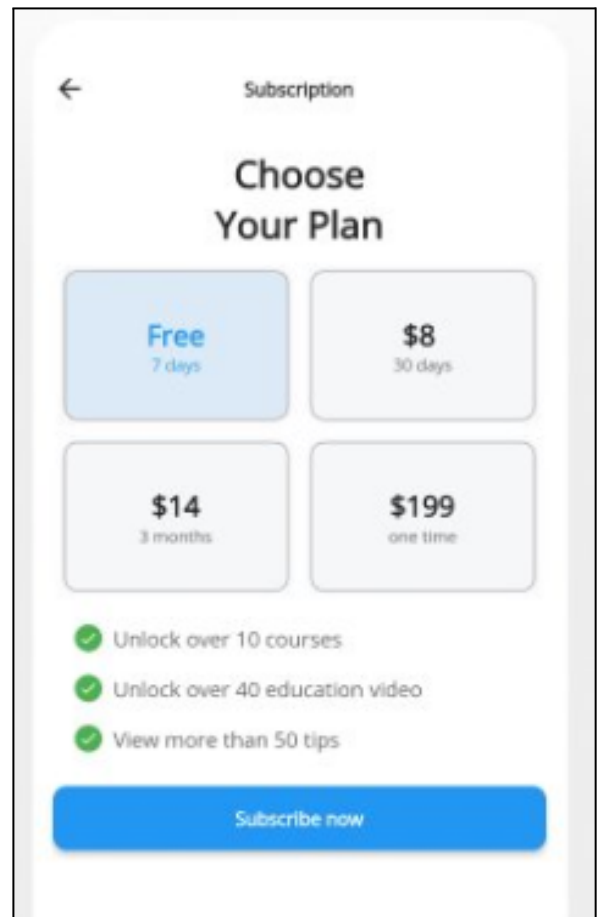


Task / Exercise C3 (Design UI in Flutter, No BL) [60 mins] :

For the following screen of a mobile application, write the flutter code for it. No need to implement any interaction or business logic. (No need for the colored bar at the left side of the card, but if you can do it, that's excellent)

Task / Exercise C4 (Events, If you are a challenger) [0 mins] :

For Exercise 1, upon clicking a box, it should be colored as the selected, whilst the previous selected cell should be colored as normal.



OPTIONAL EXERCISES

Task / Exercise 1 :

Setup your phone to wirelessly connect with your PC to debug and deploy your APK

Task / Exercise 2 :

Sign up for an account with www.FlutterFlow.io and explore their functionalities.

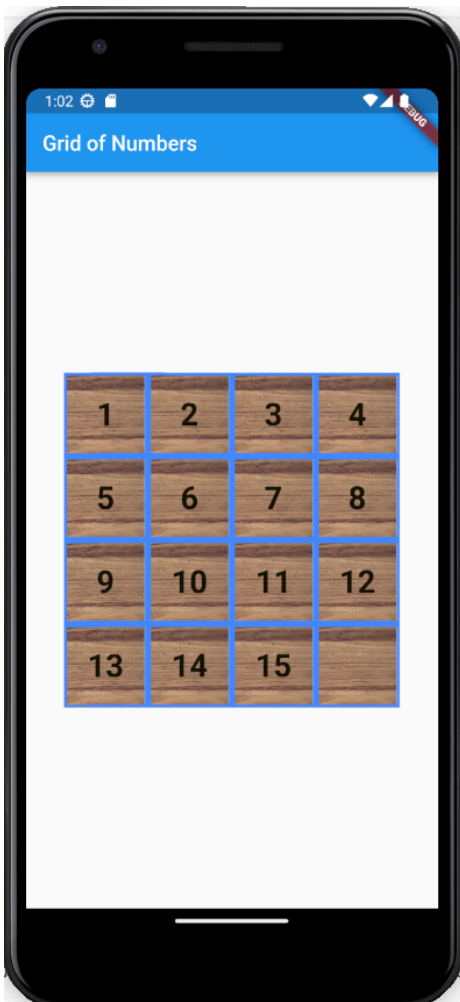
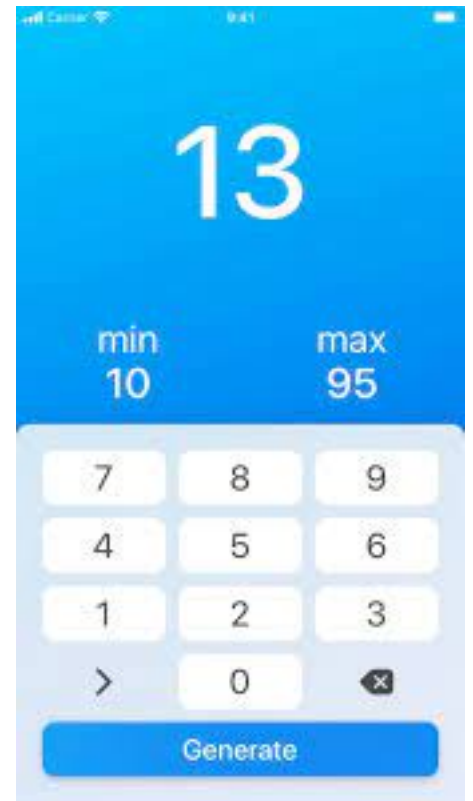
Task / Exercise 3 :

Using Flutter, create the app for :

Random Number Generator that generates random numbers within a specified range and displays them to the user.

Task / Exercise 4 :

Create a Flutter App to convert decimal numbers to binary numbers



Task / Exercise 5 :

Create a grid with numbers inside the cell as shown in the screen on the left.

Task / Exercise 6 :

Create the Flutter Widgets for the Tic Tac Toe game shown below: (**No Business Logic only, UI**)

